

MEMORANDUM

TO: Director, Game Design Certificate Program, UW-Madison

FROM: Yuzhi Duan, Graduate Student, Harvard Graduate School of Education; Alumna of UW-Madison '24, Game Design Certificate

DATE: November 11, 2025

RE: Recommendation for Adopting AI Coding Tools in Undergraduate Game Design Courses

Executive Summary

Based on my experience in the Game Design Certificate program at UW-Madison, I recommend that the program adopt a structured and iterative-focused approach to integrate AI coding tools, specifically for students with limited coding backgrounds. Rather than using AI to bypass technical learning or to produce final implementations, students should be taught to leverage AI tools strategically during prototyping and iteration phases to rapidly test design concepts. This approach preserves essential program learning objectives while addressing the significant technical barriers that currently prevent design-focused students from fully exploring their creative ideas.

Context Analysis

When I applied for the Game Design Certificate, I was excited about the creative aspects: storytelling, mechanics, and player experience. Coming from Education Studies and Communication Arts majors, I had a solid foundation in narration and media analysis, but minimal programming experience beyond the COMP SCI 200 prerequisite, where I learned only Java basics. This prepared me adequately for the core course CURRIC 357 (Game Design I), where we designed a board game and created a simple Twine story. However, CURRIC 432 (Game Design II) presented entirely different challenges. We created digital games, and while teams could choose their platforms, the technical complexity increased dramatically. My team decided to use Godot—a decision made primarily by my teammate with a computer science background. While I contributed visual art, design ideas, and narrative content, I depended on this teammate for implementation. When I wanted to test whether a particular mechanic would

create the emotional arc I envisioned, I couldn't modify the prototype myself. I had to explain my idea, wait for implementation, then evaluate the result. This gap between conceptualizing a mechanic and experiencing it in a playable form could stretch across days.

I attempted to learn Unity independently, hoping to develop more technical autonomy. I spent hours debugging code—struggles far beyond what COMP SCI 200's Java introduction had prepared me for. The collaborative workflow was valuable for learning communication and teamwork skills, which are essential for professional game development. However, it also constrained my ability to iterate rapidly on design ideas without additional coding instruction. This experience reflects a structural tension in the certificate: the program requires only one introductory computer science elective alongside courses in art, design theory, and production, expecting students will develop technical skills through formal coursework, self-directed learning, and peer collaboration. For students from social science backgrounds, this creates significant challenges when transitioning to technically demanding digital game projects.

How AI Tools Could Fit Into Game Design Education

Game design education at UW-Madison embodies key genres of learning at scale identified by Prof. Reich (2020): instructor-guided and peer-guided learning. Courses rely heavily on instructor-guided learning, though instructors focus on design thinking rather than programming. In CURRIC 277, 357, and 432, instructors guided us through design processes, facilitated playtesting, and critiqued our game concepts. They provided excellent guidance on whether our designs were compelling from player experience perspectives. The certificate also emphasized peer-guided collaborative learning. Team projects required students with varied expertise to collaborate, with instructors deliberately asking us to assign different roles based on our backgrounds. This authentically mirrors professional game development practices. Reich captures this significance: "While instructors played an important role in shaping the direction, membership, and cadence of creativity within these communities, the learning experience of each individual student in the course was dramatically shaped by the student's peer network" (2020, p. 81). Working with a teammate with a CS background taught me valuable lessons about cross-disciplinary collaboration and communication. This collaborative structure is essential to the program's pedagogical goals.

However, while role-based collaboration was valuable for learning teamwork, there were moments when I wished I could iterate more quickly on design ideas during prototyping—not to replace collaboration, but to test concepts more rapidly before bringing them to the team. Reich argues that the most promising educational technologies are "hybrids" that strategically combine elements from multiple genres to offset each genre's limitations (2020, p. 124). AI coding assistants could augment existing pedagogy by enabling faster iteration during individual prototyping phases while preserving the essential peer collaboration and role-based teamwork the program intentionally cultivates.

Recent empirical evidence strongly supports using AI tools specifically for rapid prototyping during design iteration. Alharthi's (2025) mixed-methods study of 42 game developers found that AI tools "accelerate ideation, enhance prototyping, and automate repetitive tasks"(p. 12). Industry reports correspond to this pattern: AI is already used for bug testing, dialogue creation, and other aspects of professional game development (Coursera Staff, 2025, para. 4). AI's value depends entirely on how it is positioned within the learning process. When students use AI for quick prototyping during early exploration, it may enhance creative work. When students use AI to create final implementations without understanding technical decisions, it undermines learning. Ternar et al. (2025) emphasize that current generative AI systems contribute most effectively during ideation and rapid prototyping, where speed, iteration, and low-fidelity exploration are prioritized (p. 18). When used in this capacity, AI can amplify creative experimentation and make prototyping more accessible to students with limited coding backgrounds.

In the context of the Game Design Certificate program, this evidence supports an approach that introduces AI tools as scaffolds for early-stage exploration rather than as shortcuts to production. Such use aligns with the program's dual goals of developing both technical proficiency and critical design thinking, ensuring that students gain autonomy in their creative process while still engaging deeply with the technical foundations of game development.

However, successful implementation depends not merely on tool access but on faculty ability to design what Ternar et al. (2025) term "explicit role design"—carefully structured workflows with systems positioned as creative assistants within collaborative workflows that maintain clarity

about human and machine responsibilities (p. 19). They further noted that in classroom settings, this is complemented by reports of brainstorming and knowledge gains that accelerate team progress (p.19). In other words, peer-guided learning would be enhanced by integrating AI assistance at the early stage of game design with proper instructors' guidance. This requires professional development enabling faculty to articulate when AI assistance supports learning versus when it undermines it, distinguish appropriate from inappropriate tool applications across different assignment types, and assess student learning processes rather than just artifacts.

Conclusion

The Game Design Certificate at UW–Madison already exemplifies the instructor-guided and peer-guided learning models described by Reich (2020), blending faculty mentorship with collaborative design practice. Integrating AI coding assistants can strengthen this foundation by serving as a hybrid support that enhances iteration without disrupting the program's collaborative structure. Used during early prototyping, these tools enable students, especially those with limited coding backgrounds, to test ideas more independently and bring refined concepts back to their teams for critique. Such implementation aligns with the program's learning outcomes: helping students utilize design and technical skills and discuss the critical and theoretical perspectives that shape their creative work (University of Wisconsin–Madison, n.d.). Research further shows that AI can accelerate ideation and mirror emerging professional workflows when used as a scaffold rather than a shortcut.

As a proud alumna of UW–Madison's Game Design Certificate and a current graduate student studying learning design, I offer this recommendation with deep respect for the program and its future potential. The certificate gave me the creative confidence to pursue this field, and I believe that, with the right safeguards, AI coding tools could empower the next generation of designers to experiment, reflect, and create with greater autonomy. Yet adopting such technology responsibly requires deliberate preparation and care, such as clear faculty preparation and process-based assessment. Overall, I hope this memo contributes constructively to that ongoing conversation and supports the program's continued efforts in innovative and human-centered game design education.

Reference

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